Experiment 6: Implementation of K-Means and Hierarchical Clustering

1. Dataset Source

The dataset used for this experiment is the **Iris Dataset**.

- Source: https://archive.ics.uci.edu/ml/datasets/iris
- It is a widely used dataset for clustering and classification tasks and is available publicly.

2. Dataset Description

The Iris dataset consists of measurements of iris flowers from three different species.

Features:

Feature	Description
Sepal Length	Length of sepal (cm)
Sepal Width	Width of sepal (cm)
Petal Length	Length of petal (cm)
Petal Width	Width of petal (cm)

Dataset Characteristics:

- Number of instances: 150
- Number of features: 4
- Number of clusters: 3 (based on species)
- Type: Unsupervised learning (labels not used in clustering)

3. Mathematical Formulation

🔹 **K-Means Clustering**

K-Means is a partition-based clustering algorithm that divides data into K clusters.

Objective Function:

$$
J = \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2
$$

Where:

- k = number of clusters
- C_i = set of points in cluster i
- μ_i = centroid of cluster i

The goal is to minimize the sum of squared distances between data points and their respective cluster centroids.

🔹 Hierarchical Clustering

Hierarchical clustering builds a tree-like structure called a dendrogram.

It can be of two types:

- Agglomerative (bottom-up)
- Divisive (top-down)

Distance Measures:

- Euclidean Distance
- Manhattan Distance

Linkage Methods:

- Single linkage
- Complete linkage
- Average linkage
- Ward's method

4. Algorithm Limitations

K-Means Limitations:

- Requires predefined number of clusters (K)
- Sensitive to outliers

- Assumes clusters are spherical
- May converge to local minima

Hierarchical Clustering Limitations:

- Computationally expensive for large datasets
- Difficult to decide optimal cluster number
- Once merged or split, clusters cannot be changed
- Sensitive to noise and outliers

5. Methodology / Workflow

The following steps were performed in this experiment:

1. Load the dataset
2. Select relevant numerical features
3. Visualize the dataset
4. Apply Elbow Method to determine optimal K
5. Perform K-Means clustering
6. Visualize clusters and centroids
7. Apply Hierarchical clustering
8. Plot dendrogram
9. Analyze clustering results

Workflow Diagram:

Dataset → Feature Selection → Visualization → Elbow Method
→ K-Means Clustering → Cluster Visualization
→ Hierarchical Clustering → Dendrogram → Analysis

6. Performance Analysis

- K-Means clustering performance was analyzed using the **Elbow Method**, which helps determine the optimal number of clusters by minimizing WCSS (Within Cluster Sum of Squares).
- Hierarchical clustering performance was analyzed using a **dendrogram**, which visually represents cluster formation.

Observations:

- The elbow point indicated the optimal number of clusters.
- K-Means formed well-separated clusters.
- Hierarchical clustering provided a clear hierarchical structure of data.

7. Hyperparameter Tuning

For K-Means:

- Number of clusters (K) was tuned using the Elbow Method
- Initialization method: k-means++

For Hierarchical Clustering:

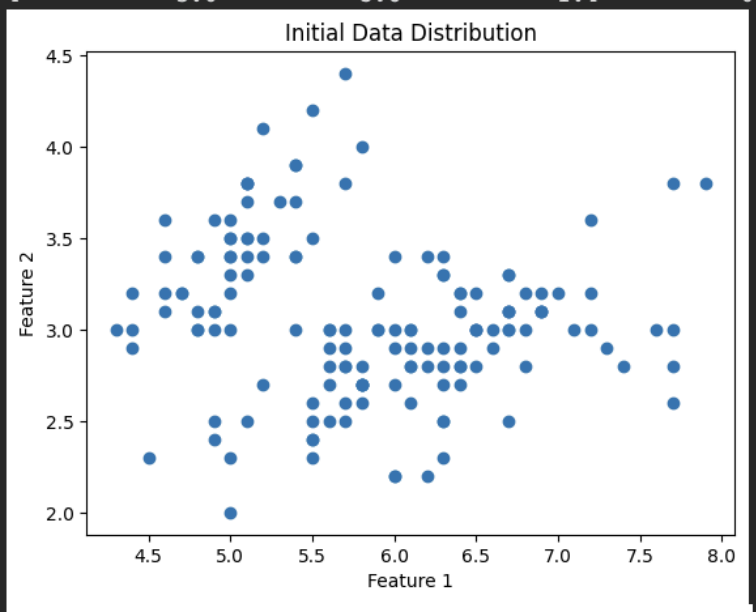
- Linkage method (Ward's method used)
- Distance metric (Euclidean distance)

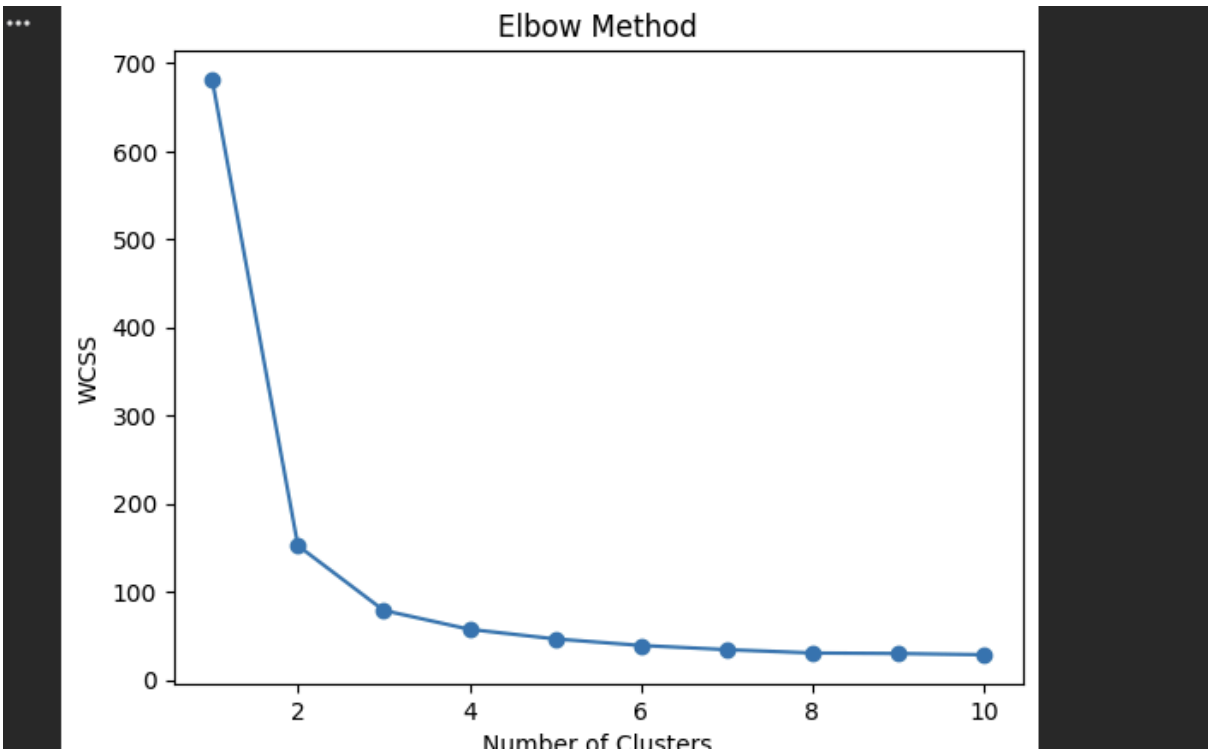
Impact:

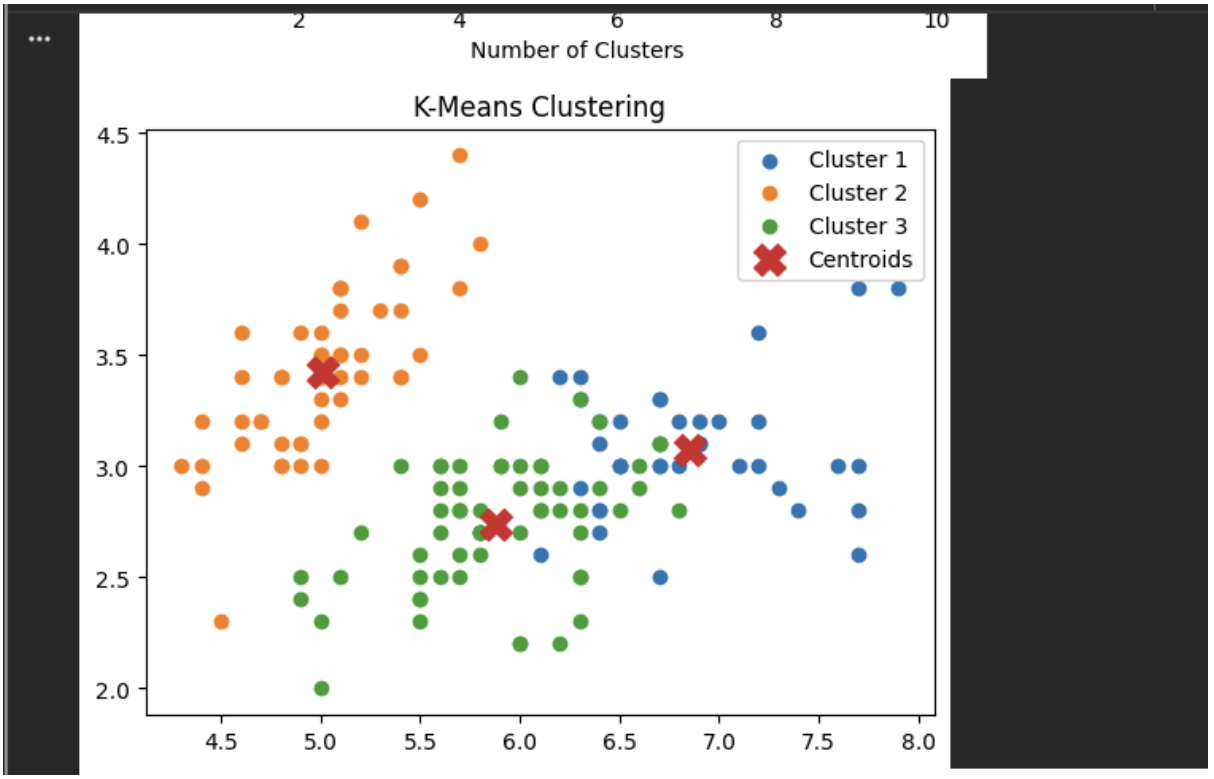
- Improved cluster quality
- Better separation of data points
- More meaningful clustering results

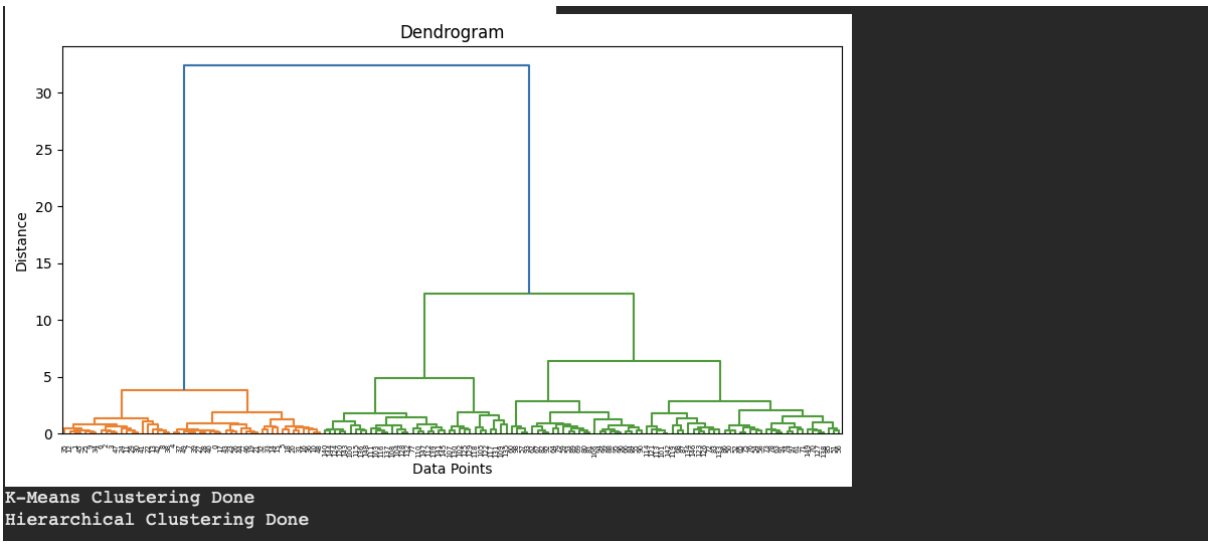
OUTPUTS:

```
...    sepal_length  sepal_width  petal_length  petal_width species
0               5.1          3.5           1.4          0.2  setosa
1               4.9          3.0           1.4          0.2  setosa
2               4.7          3.2           1.3          0.2  setosa
3               4.6          3.1           1.5          0.2  setosa
4               5.0          3.6           1.4          0.2  setosa
```




Elbow Method
WCSS
700
600
500
400
300
200
100
0
2
4
6
8
10
Number of Clusters


2
4
6
8
10
Number of Clusters
K-Means Clustering
Cluster 1
Cluster 2
Cluster 3
Centroids
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0



Conclusion

K-Means and Hierarchical clustering algorithms were successfully implemented on the dataset. K-Means efficiently grouped data into clusters based on centroid distance, while Hierarchical clustering provided a detailed dendrogram showing relationships between data points. Both methods helped in understanding the structure of the dataset, with K-Means being faster and Hierarchical clustering offering better visualization of cluster hierarchy.